
Polynomial-Linked Tucker Models for Nonlinear Low-Rank Tensor Observations

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Abstract

1 Classical Tucker models explain multiway data through multilinear low-rank struc-
2 ture, but many measurement processes introduce nonlinear entry-level effects such
3 as saturation, response curves, and material-dependent nonlinearities. We study
4 a focused question: can one add nonlinear observation capacity to Tucker while
5 preserving its parameter-efficient low-rank backbone?

6 We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL),
7 which applies a finite polynomial link to a shared Tucker latent tensor. The model
8 contains Tucker as a linear special case, acts as a compressed surrogate for high-
9 order Tucker interactions, and admits masked learning through alternating gradient
10 updates and a small ridge-regression subproblem for the link coefficients. We give
11 approximation and separation results, together with a masked-entry generalization
12 bound. Experiments on controlled tensors, CAVE hyperspectral completion and
13 reconstruction, and external HSI scenes show that NTD-PL improves matched-rank
14 Tucker when a shared nonlinear residual is present, that the effect is not explained
15 by modest rank increases or post-hoc scalar calibration, and that boundary cases
16 arise when Tucker already leaves little coherent residual for the link to capture.

1 Introduction

18 Tensor decompositions provide compact models for multiway data in scientific sensing, imaging,
19 and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker
20 decomposition is especially useful because it separates a small core from mode-wise factors, but
21 its observation model is linear in the reconstructed latent entry. This is a strong assumption in
22 measurement pipelines where latent signals may pass through saturation, radiometric response curves,
23 material-dependent nonlinearities, or other entry-level generation effects. After such a response, an
24 intrinsically low-rank latent tensor can appear substantially higher-rank to a linear Tucker model.

25 One way to absorb this mismatch is to increase Tucker rank, but doing so expands the core and factor
26 dimensions everywhere in the model. Another is to calibrate the fitted Tucker output post hoc, but
27 that leaves the learned latent geometry fixed. We ask a narrower question: can a small nonlinear
28 observation model, learned jointly with the Tucker factors, make a tight multilinear rank budget more
29 effective? This paper focuses on the case where the residual is well described by a shared entry-level
30 response. The goal is therefore not to replace rank selection or general nonlinear tensor decoders, but
31 to isolate a simple mechanism by which nonlinear observation effects can be represented inside a
32 Tucker backbone.

33 We instantiate this idea with a polynomial link over a multilinear low-rank latent tensor. Given a
34 Tucker latent tensor

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad (1)$$

we model the observed tensor by a polynomial link

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}. \quad (2)$$

We call this model NTD-PL. It preserves the Tucker backbone and adds only $P + 1$ scalar link coefficients, while the powers $\mathcal{S}^{\odot p}$ induce tied high-order interactions among the same latent factors. Thus the link is not an independent high-rank correction; it is a structured, parameter-efficient way to expose nonlinear tensor structure already organized by the Tucker factors.

Contributions. We make four contributions.

1. We introduce NTD-PL, a Tucker model with a jointly learned polynomial observation link, with standard Tucker recovered as the degree-one special case.
2. We show that polynomial links act as compressed high-order Tucker interactions, yielding approximation and separation results for nonlinear low-rank observations.
3. We give a masked learning objective, a practical alternating algorithm with degree continuation, and a masked-entry generalization bound.
4. We empirically test the mechanism rather than a broad benchmark claim: controlled tensors isolate when the link should help, held-out CAVE completion tests low-rank generalization, reconstruction checks separate the effect from rank-lift and post-hoc calibration, and external HSI scenes show both transfer and boundary cases.

2 Related Work

Low-rank tensor models. Classical tensor decompositions such as CP, Tucker, tensor train, and tensor ring provide compact representations for multiway data [Kolda and Bader, 2009, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for our work because its core tensor and mode factors give a flexible multilinear backbone. Most methods in this family improve the representation by changing ranks, factor structures, regularizers, or optimization procedures. NTD-PL keeps the same Tucker parameterization and instead changes the observation map from the latent tensor to the measured entries.

Nonlinear tensor factorization. Prior work has introduced nonlinearity through generalized likelihoods, kernels, Gaussian processes, neural decoders, and nonlinear tensor network architectures [Hong et al., 2020, Xu et al., 2011, Zhe et al., 2016, Pan et al., 2020, Saragadam et al., 2024, Varol-gunes et al., 2023]. These approaches are often more expressive than polynomial links, but they can trade off structural transparency, optimization simplicity, or explicit finite-dimensional analysis. Our goal is complementary. We use a minimal shared observation link so that the nonlinear effect can still be analyzed as a structured extension of Tucker rather than as an unconstrained decoder.

Polynomial feature lifts and high-order interactions. Polynomial expansions and Volterra-style models are classical tools for representing nonlinear systems. In tensor settings, high-order interactions can be represented explicitly but often incur rapidly growing parameter and contraction costs. Polynomial matrix and tensor factorization methods use feature expansions to increase expressivity, but they typically introduce larger lifted spaces, independent high-order parameters, or model-specific factor structures. NTD-PL can be viewed as a shared-backbone compression of high-order Tucker interactions: the lifted features are induced by powers of one learned latent tensor, not by an independent high-order tensor parameter.

Rank lifting and post-hoc calibration. Two natural alternatives explain why the empirical comparisons are organized as mechanism checks. Rank lifting gives Tucker more ordinary multilinear capacity and can absorb part of the nonlinear residual, but it enlarges the core and factors throughout the model. Post-hoc scalar calibration is cheaper, but it remaps a Tucker reconstruction after the latent geometry has already been learned. NTD-PL sits between these alternatives: it adds only a small shared link, but learns that link jointly with the Tucker factors.

80 3 Polynomial-Linked Tucker Models

81 3.1 Model and Observation Assumption

82 Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ be an observed tensor and let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad \mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \quad (3)$$

83 NTD-PL treats $\mathcal{S}$ as a shared low-rank latent tensor and models each observed entry through a scalar
84 polynomial response:

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad f_{\beta}(s) = \sum_{p=0}^P \beta_p s^p, \quad \mathbf{i} = (i_1, \dots, i_N). \quad (4)$$

85 Equivalently, the tensor prediction is

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad \beta \in \mathbb{R}^{P+1}. \quad (5)$$

86 The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel,
87 and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is
88 recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the entry-generation map
89 while leaving the Tucker backbone and its multilinear ranks fixed.

90 3.2 Shared-Backbone High-Order Interactions

91 Although f_{β} is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. Expanding one
92 entry gives

$$S_{\mathbf{i}}^p = \left(\sum_{r_1=1}^{R_1} \dots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)} \right)^p \quad (6)$$

$$= \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}. \quad (7)$$

93 Thus a degree- p channel couples p latent Tucker index tuples while reusing the same mode factors
94 in every copy. This is the sense in which the polynomial link acts as a shared-backbone high-order
95 Tucker interaction: it forms the kind of products that would appear in an explicit p -fold Tucker
96 expansion, but ties all mode projections to $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}$ and mixes the resulting channels only
97 through the scalar coefficients $\{\beta_p\}$.

98 This distinction also separates NTD-PL from post-hoc calibration. A post-hoc calibrator learns a
99 scalar map after a Tucker tensor has already been fitted. NTD-PL instead learns the latent tensor
100 and the link jointly, so the nonlinear response can shape the low-rank geometry itself. Section 4
101 formalizes the induced rank expansion and the resulting separation from standard Tucker.

102 3.3 Masked Least-Squares Objective

103 Let $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_{\mathbf{i}} \Omega_{\mathbf{i}}$. The empirical
104 data-fitting term is the average squared residual over observed entries,

$$\mathcal{L}_{\Omega} = \frac{1}{2|\Omega|} \sum_{\mathbf{i}: \Omega_{\mathbf{i}}=1} (Y_{\mathbf{i}} - f_{\beta}(S_{\mathbf{i}}))^2. \quad (8)$$

105 The squared loss is the standard least-squares choice for real-valued reconstruction and completion,
106 and it aligns with the RMSE-based evaluation used in the experiments. The Frobenius form below is
107 simply the tensor notation for the same observed-entry residual sum:

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2. \quad (9)$$

The factor $1/|\Omega|$ keeps the loss and gradient scale comparable across full-observation reconstruction and different missing rates, while the factor $1/2$ removes a constant 2 from the residual gradients. The Frobenius penalties are ridge/Tikhonov regularizers for the core and mode factors; the ℓ_2 penalty on β stabilizes the polynomial link and makes the coefficient subproblem well-conditioned. When $\Omega \equiv \mathbf{1}$, the same objective gives full tensor reconstruction; otherwise it gives masked tensor completion.

Practical scaling. Like Tucker, NTD-PL has scale indeterminacies between the core and mode factors. Polynomial powers make this numerically more sensitive: if the scale of $\mathcal{S}$ drifts, different powers change at different rates and the link coefficients become poorly conditioned. In the implementation we initialize from Tucker, normalize factor scales during training, and absorb the inverse scales into the core. This gauge fixing does not change the latent tensor $\mathcal{S}$ or the model class, but it makes the alternating optimization more stable in the low-rank regimes used in the experiments.

4 Theory

The analysis formalizes the mechanism behind NTD-PL: polynomial links preserve the low-rank Tucker latent tensor while inducing structured high-order interactions in the observed tensor. The results below are representation results. They explain what the model class can express under a shared entrywise observation mechanism; they do not assert identifiability or guarantee that a nonconvex optimizer recovers the representing parameters.

4.1 Approximation Under Nonlinear Observations

Theorem 1 (Polynomial generation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P . Then degree- P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 2 (Analytic link approximation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and $\|\mathcal{S}^*\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree- P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \|\mathcal{Y} - \hat{\mathcal{Y}}\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \quad M = \prod_{n=1}^N I_n. \quad (10)$$

This is the normalized Frobenius error, matching the scale of RMSE-style reconstruction metrics. The bound is intentionally stated for the simple Taylor truncation; stronger best-polynomial bounds can be substituted when additional regularity or a specific approximation basis is desired.

4.2 Rank Expansion and Separation

Proposition 1 (Shared-backbone rank expansion). *Let $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ have Tucker rank at most $(R_1, \dots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (11)$$

More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{(p)} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by

$$A_{i,\alpha}^{(p)} = \prod_{r=1}^R A_{i,r}^{\alpha_r}, \quad (12)$$

where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)}$.

The entrywise expansion in Section 3 can therefore be regrouped as an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same backbone factors. This is the mechanism behind the separation below: $\mathcal{S}^{\odot p}$ is not a free high-rank correction, but a tied high-order interaction generated from one latent tensor.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω , rank $(R_1, \dots, R_N)$, degree P

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1: initialize  $\mathcal{X}, \{\mathbf{A}^{(n)}\}$  from Tucker on the masked tensor
2: initialize active degree  $P_{\text{act}} \leftarrow 1$  and  $\beta = (0, 1)$ 
3: for  $t = 1, \dots, T$  do
4:   form  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ 
5:   if a link update is scheduled then
6:     update active coefficients  $\beta_{0:P_{\text{act}}}$  by ridge regression on  $\{(S_i, Y_i) : \Omega_i = 1\}$ 
7:   end if
8:   compute  $f_\beta(\mathcal{S})$  and  $f'_\beta(\mathcal{S})$ 
9:   take first-order steps on  $\mathcal{X}$  and  $\{\mathbf{A}^{(n)}\}$  using the masked gradient
10:  normalize factor scales and absorb them into the core
11:  increase  $P_{\text{act}}$  according to the continuation schedule, initializing the new coefficient at zero
12: end for
13: return  $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta$ 

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Theorem 3 (Tensorial separation from standard Tucker). *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose $I_n \geq D$ for every n , where $D = \binom{R+p-1}{p}$. Then there exists an order- N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ that is representable by a degree- p NTD-PL model with Tucker backbone rank $(R, \dots, R)$, but whose exact standard Tucker rank is*

$$(D, \dots, D). \quad (13)$$

Equivalently, a degree- p polynomial link can generate tensors whose multilinear ranks are $((\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p}))$ from a shared Tucker backbone of rank $(R, \dots, R)$.

The construction, given in Appendix A, uses a CP-diagonal Tucker backbone and the fact that generic Veronese feature matrices have full column rank. The theorem does not say that NTD-PL represents arbitrary high-rank Tucker tensors. Rather, it identifies a structured high-rank Tucker subset generated by a low-rank shared backbone. This is the precise sense in which the polynomial link provides a tied rank lift rather than an independent high-rank correction.

Together, these results give the intended scope of the model. If the data are generated by a shared entrywise nonlinear response, polynomial links give a finite-dimensional approximation route on top of the same latent tensor. At the same time, the powers of that latent tensor induce Veronese-lifted Tucker factors, explaining why a scalar link can change multilinear rank structure without introducing an arbitrary high-rank tensor parameter.

5 Optimization

The objective in Section 3 is nonconvex in the Tucker core, factors, and link coefficients jointly, but it has a useful block structure. With the polynomial link fixed, the loss can be differentiated through the Tucker reconstruction. With the latent tensor fixed, the link coefficients solve a low-dimensional ridge regression on the currently observed entries. Algorithm 1 summarizes the solver used in the experiments.

Unlike a post-hoc calibration baseline, the Tucker tensor is not held fixed after initialization. The link update uses the current latent tensor, and the subsequent backbone gradient step is taken under the current nonlinear link. This coupling is the optimization counterpart of the model distinction made in Section 3.

5.1 Backbone Gradient Step

For a fixed β , define

$$f'_\beta(\mathcal{S}) = \sum_{p=1}^P p \beta_p \mathcal{S}^{\odot(p-1)}. \quad (14)$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - \mathcal{Y}) \odot f'_\beta(\mathcal{S}). \quad (15)$$

176 Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \cdots \times_N \mathbf{A}^{(N)\top} + \lambda_{\mathcal{X}} \mathcal{X}. \quad (16)$$

177 For the n -th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \cdots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \cdots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \quad (17)$$

178 Then the factor gradient can be written in unfolding form as

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \quad (18)$$

179 The experiments use first-order updates for these gradients. The algorithmic claim does not depend
180 on a particular adaptive optimizer; the important point is that the backbone is updated jointly with the
181 current nonlinear link.

182 5.2 Closed-Form Link Update

183 With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is ridge regression. Let $\mathbf{y}_{\Omega}$
184 collect the observed values and let $\Phi_{\Omega, Q}(\mathcal{S})$ have rows $[1, s, s^2, \dots, s^Q]$ for $s = S_i$, $\Omega_i = 1$. The
185 subproblem corresponding to the normalized objective in Section 3 is

$$\min_{\beta_{0:Q}} \frac{1}{2|\Omega|} \|\mathbf{y}_{\Omega} - \Phi_{\Omega, Q}(\mathcal{S}) \beta_{0:Q}\|_2^2 + \frac{\lambda_{\beta}}{2} \|\beta_{0:Q}\|_2^2. \quad (19)$$

186 Hence

$$\beta_{0:Q}^* = (\Phi_{\Omega, Q}^{\top} \Phi_{\Omega, Q} + |\Omega| \lambda_{\beta} \mathbf{I})^{-1} \Phi_{\Omega, Q}^{\top} \mathbf{y}_{\Omega}. \quad (20)$$

187 Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal
188 equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In
189 practice the least-squares system is tiny because $Q \leq P \leq 6$ in our experiments.

190 5.3 Continuation and Stabilization

191 The polynomial family is nested in the maximum degree, so we use a continuation schedule that
192 starts from the Tucker-compatible degree-one model and activates one additional degree at a time:

$$t_k = \text{round} \left(\frac{kT}{P} \right), \quad k = 1, \dots, P-1.$$

193 When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new
194 coefficient β_{p+1} is initialized at zero. The model therefore expands from a linear low-rank fit instead
195 of starting from a high-degree polynomial link immediately.

196 This staging is useful because high powers of $\mathcal{S}$ are sensitive to scale and can dominate early gradients.
197 We also normalize factor columns during training and absorb their scales into the core; this is a
198 gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For numerical stability in the
199 ridge update, the implementation may solve the Vandermonde system on a centered and scaled latent
200 variable $(S - \mu)/\sigma$, then convert the coefficients back to the original power basis of $\mathcal{S}$. This changes
201 only the conditioning of the coefficient solve, not the model being optimized.

202 5.4 Cost

203 The dominant cost is forming the Tucker reconstruction and its reverse contractions on the observed
204 tensor. For dense tensors this is comparable to standard Tucker optimization at the same rank,
205 plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms a
206 Vandermonde design or the corresponding moment equations and solves a $(P+1) \times (P+1)$ linear
207 system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the
208 Tucker contractions. Thus the polynomial link changes the observation model without introducing a
209 tensor-valued high-order parameter block.

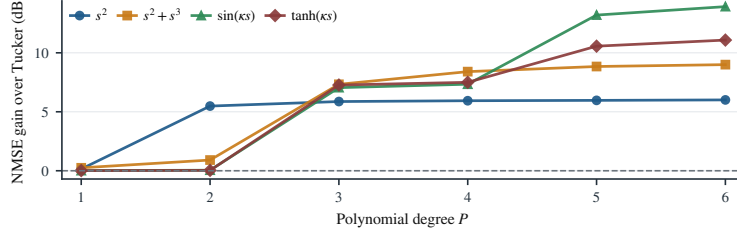


Figure 1: Controlled nonlinear tensors. NTD-PL gains over Tucker increase as orthogonal nonlinear residual energy increases.

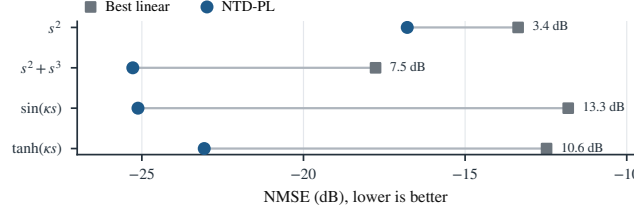


Figure 2: Endpoint controlled-tensor comparison. Each row compares the best linear baseline with NTD-PL on normalized error; horizontal distance is the gain in dB.

6 Experiments

The empirical study tests a scoped mechanism: when a Tucker backbone is kept low-rank, can a shared polynomial link capture residual structure that is invisible to a linear Tucker observation model? The experiments are organized to separate this question from a broad benchmark claim. Controlled tensors isolate when the link should help, masked CAVE completion tests whether the learned low-rank geometry generalizes to held-out entries, CAVE reconstruction ablations rule out simple rank-lift and post-hoc calibration explanations, and external HSI scenes probe transfer and boundary cases.

Protocol. Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. NTD-PL is compared with Tucker at the same multilinear rank, except in the explicit rank-lift ablation where Tucker is given extra rank. We report RMSE and spectral angle (SAM), with NMSE(dB) used for cross-domain reconstruction where scale varies across tensors. Completion masks entries uniformly at random and computes all completion metrics only on the held-out entries.

6.1 Controlled Nonlinear Tensors

The first check isolates the intended effect. We generate tensors of the form $Y_\alpha = \sqrt{1-\alpha}S^* + \sqrt{\alpha}\tilde{R}^*$, where S^* is low-rank Tucker and $\tilde{R}^*$ is the normalized component of a candidate nonlinear response that is orthogonal to S^* . Thus α controls nonlinear residual energy rather than the amplitude of a raw scalar nonlinearity. When the source is linear, NTD-PL reduces to Tucker up to the fitted affine link; when nonlinear residual energy is present, the higher-order link terms should become useful. Figure 1 shows exactly this pattern: NTD-PL matches Tucker on linear sources and improves as the nonlinear residual grows. Figure 2 summarizes the endpoint comparison against the best linear baseline. The formal residual construction, full alpha sweep, polynomial-degree sweep, degree-continuation training curves, and fitted link-term contributions are reported in Appendix B.1.

6.2 Low-Rank Masked Completion

Masked completion is the main empirical test because held-out entries expose whether the learned low-rank geometry transfers beyond the observed tensor. In this setting, the polynomial link is useful only if it changes the inductive bias of the shared Tucker backbone, rather than simply calibrating training entries. Table 1 shows that this effect persists across random-mask rates at full resolution: with the same rank budget as Tucker, NTD-PL gives consistent held-out RMSE gains and also

Table 1: Full-resolution CAVE completion at rank (33, 33, 4) under random masks. Metrics are computed on missing entries. Tucker and NTD-PL columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over Tucker.

ρ	RMSE* $\downarrow$			SAM* $\downarrow$		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 $\pm$ 0.0179	0.0229$\pm$0.0158	12.9%	11.82 $\pm$ 5.33	8.36$\pm$3.38	29.3%
0.3	0.0263 $\pm$ 0.0179	0.0230$\pm$0.0158	12.5%	14.75 $\pm$ 6.33	12.47$\pm$5.06	15.5%
0.5	0.0264 $\pm$ 0.0179	0.0230$\pm$0.0158	12.9%	15.56 $\pm$ 6.85	13.48$\pm$5.75	13.4%
0.7	0.0266 $\pm$ 0.0180	0.0219$\pm$0.0128	17.7%	16.02 $\pm$ 7.21	13.79$\pm$5.43	13.9%

Table 2: CAVE full reconstruction at matched ranks. Metrics are averaged over 15 scenes.

Rank	Params	RMSE $\downarrow$		SAM $\downarrow$		Gain	
		Tucker	NTD-PL	Tucker	NTD-PL	RMSE	SAM
(18, 18, 3)	20k	0.0369	0.0319	22.36	19.23	13.6%	14.0%
(24, 24, 4)	27k	0.0299	0.0256	17.42	15.15	14.3%	13.0%
(33, 33, 4)	38k	0.0261	0.0223	16.43	14.41	14.3%	12.3%
(40, 40, 5)	49k	0.0217	0.0192	12.65	11.85	11.5%	6.4%

improves spectral angle on most scenes. The scene-level sign and Wilcoxon tests in Appendix B.2 support the same conclusion.

6.3 CAVE Reconstruction Mechanism Checks

Full-observation reconstruction is a controlled mechanism check rather than the main generalization test: with all entries observed, we can ask whether the polynomial link consistently removes residual structure left by a matched-rank Tucker backbone. Table 2 supports this interpretation. NTD-PL improves both intensity error and spectral angle across the rank range, while the advantage becomes smaller as the Tucker rank increases. This trend is the expected signature of a lightweight observation model: it helps most when the linear backbone is tight, and becomes less necessary as ordinary multilinear capacity grows.

Two ablations clarify what the gain means. First, Table 3 asks how much spatial Tucker rank is needed before ordinary Tucker first matches the RMSE of a lower-rank NTD-PL model, while holding the spectral rank fixed. Tucker can buy back part of the RMSE gap by increasing rank, so NTD-PL should not be read as a replacement for rank selection. The spectral-angle gap, however, remains visible at the first RMSE-matching Tucker ranks, suggesting that the link is not merely trading for a slightly larger multilinear model. Second, post-hoc scalar calibration of a fitted Tucker reconstruction improves Tucker but remains below joint fitting in RMSE and SAM; the comparison and implementation details are given in Appendix B.3.

6.4 External HSI Validity and Boundary Cases

Table 4 asks whether the CAVE mechanism transfers to other hyperspectral tensors and where the shared-link assumption weakens. NTD-PL improves reconstruction and completion on Jasper Ridge, Samson, and Urban. Cuprite is the boundary case: Tucker is already strong, leaving little coherent residual for a shared polynomial link. Thus the evidence supports a scoped claim: NTD-PL is most useful under a tight rank budget when the linear Tucker residual is coherent enough to be modeled by a shared nonlinear link. Additional cross-domain checks are in Appendix B.5.

Table 3: Rank-lifted Tucker comparison on CAVE reconstruction. The table reports the first fixed-channel Tucker rank that matches each NTD-PL RMSE.

NTD-PL reference				First fixed-channel Tucker matching RMSE				Extra
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
(18, 18, 3)	20k	0.0319	19.23	(28, 28, 3)	31k	0.0316	21.41	+59.5%
(24, 24, 4)	27k	0.0256	15.15	(35, 35, 4)	41k	0.0254	16.26	+51.3%
(33, 33, 4)	38k	0.0223	14.41	(49, 49, 4)	60k	0.0222	15.28	+56.5%

Table 4: External HSI reconstruction and completion. Columns report RMSE; gains are relative RMSE reductions over Tucker, and positive Δ SAM favors NTD-PL.

Dataset	Reconstruction				Completion			
	Tucker	NTD-PL	Gain	Δ SAM	Tucker	NTD-PL	Gain	Δ SAM
Jasper Ridge	0.0462	0.0430	6.93%	0.98	0.0471	0.0434	7.82%	1.16
Samson	0.0263	0.0242	7.95%	0.06	0.0268	0.0244	8.86%	0.31
Urban	0.0444	0.0429	3.59%	0.29	0.0454	0.0433	4.67%	0.49
Cuprite	0.0176	0.0177	-0.60%	-0.04	0.0179	0.0178	0.56%	-0.00

7 Limitations

NTD-PL assumes that the residual left by a low-rank Tucker backbone is well described by a shared entry-level nonlinear response. It is therefore not a universal tensor model: when Tucker already fits well at the chosen rank, or when residual nonlinearities vary strongly across space, spectra, or modes, a single scalar link may add little.

The method also does not remove rank selection. Increasing Tucker rank can recover part of the RMSE gap, and NTD-PL inherits Tucker’s nonconvexity and initialization sensitivity. Although the link adds few parameters, evaluating it over all observed entries still increases runtime relative to plain Tucker.

8 Conclusion

We introduced NTD-PL, a Tucker model with a shared polynomial observation link. Rather than replacing multilinear low-rank structure, NTD-PL adds a small nonlinear layer that expresses structured residuals left by a linear Tucker fit. The theory explains this as tied high-order Tucker interactions, or a structured rank lift without an independent high-rank correction.

The experiments support the same scoped interpretation. Controlled tensors, held-out CAVE completion, reconstruction mechanism checks, and external HSI scenes show that the link helps when a shared nonlinear residual is plausible under a tight rank budget, while also revealing near-linear boundary cases. Adaptive links and broader tensor modalities are natural next steps.

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314 A Additional Proofs

315 A.1 Polynomial and Analytic Approximation

316 *Proof of Theorem 1.* Since g has degree at most P , write

$$g(s) = \sum_{p=0}^P c_p s^p. \quad (21)$$

317 Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^*$, and set $\beta_p = c_p$. Then, entrywise,

$$\hat{\mathcal{Y}}_{i_1, \dots, i_N} = \sum_{p=0}^P \beta_p (\mathcal{S}_{i_1, \dots, i_N}^*)^p = g(\mathcal{S}_{i_1, \dots, i_N}^*) = \mathcal{Y}_{i_1, \dots, i_N}. \quad (22)$$

318 Thus $\hat{\mathcal{Y}} = \mathcal{Y}$. □

319 **Lemma 1** (Scalar-to-tensor approximation). *Let $\|\mathcal{S}^*\|_\infty \leq B$. For any scalar function g and any*
320 *degree- P polynomial q_P ,*

$$\|g(\mathcal{S}^*) - q_P(\mathcal{S}^*)\|_F \leq \sqrt{\prod_{n=1}^N I_n} \sup_{|s| \leq B} |g(s) - q_P(s)|. \quad (23)$$

321 *Proof.* Every entry of $\mathcal{S}^*$ lies in $[-B, B]$, so every entrywise approximation error is bounded by
322 $\sup_{|s| \leq B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the
323 claim. □

324 *Proof of Theorem 2.* Let $g(s) = \sum_{k=0}^\infty a_k s^k$ be the Taylor expansion of g around zero, and choose
325 $q_P(s) = \sum_{k=0}^P a_k s^k$. By Cauchy’s estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \quad (24)$$

326 For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k| |s|^k \quad (25)$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \quad (26)$$

327 Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^*$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized
 328 bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^N I_n$, gives the normalized form used in Theorem 2. $\square$

329 A.2 Rank Expansion

330 *Proof of Proposition 1.* Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1, \dots, i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (27)$$

331 Taking an element-wise p -th power gives

$$(\mathcal{S}_{i_1, \dots, i_N})^p = \prod_{q=1}^p \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \prod_{n=1}^N A_{i_n, r_n^{(q)}}^{(n)} \right) \quad (28)$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \left(\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \right), \quad (29)$$

332 where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n , let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n ,
 333 so $|\alpha_n| = p$. Then

$$\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} = \prod_{r=1}^{R_n} \left(A_{i_n, r}^{(n)} \right)^{\alpha_{n,r}} = A_{i_n, \alpha_n}^{(n)\langle p \rangle}. \quad (30)$$

334 Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define
 335 the aggregated core

$$\mathcal{X}_{\alpha_1, \dots, \alpha_N}^{(p)} = \sum_{\substack{\gamma_1, \dots, \gamma_N: \\ \text{count}(\gamma_n) = \alpha_n \forall n}} \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}}. \quad (31)$$

336 The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = \llbracket \mathcal{X}^{(p)}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle} \rrbracket. \quad (32)$$

337 The n -th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

338 A.3 Separation

339 **Lemma 2** (Full-rank Veronese evaluations). *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$*
 340 *whose Veronese feature matrix $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D \times D}$ has rank D . Consequently, for $I \geq D$, a generic*
 341 *matrix $\mathbf{A} \in \mathbb{R}^{I \times R}$ has $\text{rank}(\mathbf{A}^{\langle p \rangle}) = D$.*

342 *Proof.* The homogeneous degree- p monomials in R variables are linearly independent polynomials.
 343 Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is
 344 nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a
 345 nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the
 346 first claim.

347 For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix
 348 is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is
 349 therefore a generic property, so $\mathbf{A}^{\langle p \rangle}$ has full column rank for generic $\mathbf{A}$. $\square$

350 *Proof of Theorem 3.* Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in$
 351 $\mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}. \quad (33)$$

352 Its mode- n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)} \right)^\top. \quad (34)$$

353 For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R , and the Khatri-Rao product on the right has rank R . Thus
 354 $\text{rank}(\mathcal{S}_{(n)}) = R$ for every n , so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

355 Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree- p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1, \dots, i_N} = \left(\sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p \quad (35)$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left(A_{i_n, r}^{(n)} \right)^{\alpha_r}, \quad (36)$$

356 where $c_\alpha = p! / (\alpha_1! \dots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full
 357 column rank D for generic $\mathbf{A}^{(n)}$. The mode- n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \text{diag}(c) \left(\mathbf{B}^{(N)} \odot \dots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \dots \odot \mathbf{B}^{(1)} \right)^\top. \quad (37)$$

358 The diagonal matrix $\text{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices
 359 is full column rank, the right factor has rank D . Therefore $\text{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every
 360 mode n , and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

361 A.4 Masked Generalization

362 **Theorem 4** (Masked-entry uniform convergence). *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set*
 363 *with*

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P+1).$$

364 *For $\theta \in \Theta$, define the full-entry risk*

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \quad M = \prod_{n=1}^N I_n,$$

365 *and the masked empirical risk*

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta),$$

366 *where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G -Lipschitz*
 367 *in θ , uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ , then with probability at*
 368 *least $1 - \delta$,*

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

369 *for every $0 < \varepsilon \leq B_\Theta$.*

370 *Proof of Theorem 4.* Fix $0 < \varepsilon \leq B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε -net of Θ in Euclidean norm. Since Θ lies
 371 in a d -dimensional ball of radius B_Θ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \quad (38)$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{1t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr \left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \geq u \right) \leq 2 \exp \left(-\frac{2mu^2}{B_\ell^2} \right). \quad (39)$$

Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1 - \delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \leq B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \quad (40)$$

Substituting the net-size bound yields the stated stochastic term.

Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \leq \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon. \quad (41)$$

Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \quad (42)$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \quad (43)$$

$$\leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \quad (44)$$

Taking the supremum over $\theta \in \Theta$ completes the proof. $\square$

Remark 1. The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on P , but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.

B Additional Experiments

B.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 5 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^*)$. Let S^* be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^* = g(S^*) - \frac{\langle g(S^*), S^* \rangle}{\|S^*\|_F^2} S^*, \quad (45)$$

and, when $\|R^*\|_F > 0$, normalize it as

$$\widetilde{R}^* = \frac{\|S^*\|_F}{\|R^*\|_F} R^*. \quad (46)$$

The observed tensor is then

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \widetilde{R}^*, \quad 0 \leq \alpha \leq 1. \quad (47)$$

By construction, $\langle S^*, \widetilde{R}^* \rangle = 0$ and $\|\widetilde{R}^*\|_F = \|S^*\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2$, $g(s) = s^2 + s^3$, $g(s) = \sin(\kappa s)$, and $g(s) = \tanh(\kappa s)$, with κ set from the input scale.

Table 5: Linear consistency on synthetic Tucker tensors. The $p > 1$ columns report effective higher-order NTD-PL contribution.

Method	bias = 0			bias = 0.5		
	RMSE↓	NMSE(dB)↓	$p > 1$	RMSE↓	NMSE(dB)↓	$p > 1$
Tucker	0.0114	-38.88	—	0.0647	-24.05	—
NTD-PL, $\beta_0 = 0$	0.0116	-38.72	4.19%	0.0522	-25.85	39.17%
NTD-PL, β_0 free	0.0117	-38.65	3.19%	0.0118	-38.58	2.98%

Table 6 shows a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Figures 3 and 4 give the fuller nonlinear sweep behind the compact main-text figures. The alpha sweep shows that NTD-PL separates from linear tensor models as nonlinear residual energy increases; the degree sweep shows that this gain is tied to the expressive order of the polynomial link rather than to an arbitrary post-processing effect.

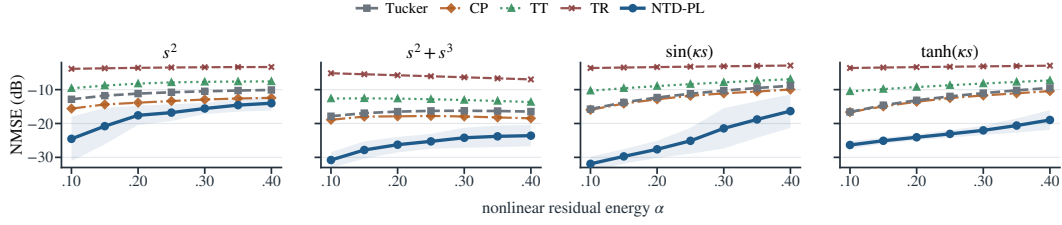


Figure 3: Controlled nonlinear alpha sweep. Each panel uses a different candidate response to define the orthogonal residual; lower NMSE(dB) is better.

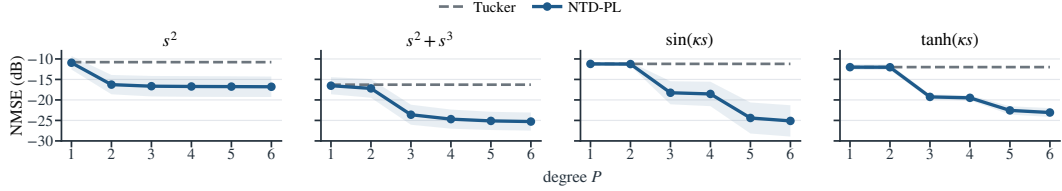


Figure 4: Controlled nonlinear degree sweep. Increasing the polynomial degree helps when the residual is induced by higher-order structure.

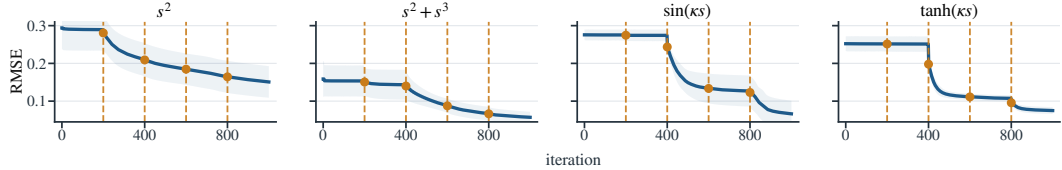


Figure 5: NTD-PL training curves for the controlled nonlinear mechanism checks. Dashed vertical lines mark degree-activation events.

407 B.2 CAVE Completion Significance and Robustness

408 The main text reports the full-resolution random-missing CAVE completion table at rank $(33, 33, 4)$.
 409 We do not repeat that table here; instead, Table 7 gives the paired sign and Wilcoxon tests for the
 410 same protocol.

411 B.3 Post-Hoc Calibration Implementation

412 The calibration baselines in Figure 6 are post-processing models applied to a fixed Tucker recon-
 413 struction. Let $\hat{\mathcal{Y}}_T$ denote the Tucker output and $\mathcal{Y}$ the target tensor. PolyCal fits a scalar degree-four

Table 6: Representative effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at the matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
s^2	0.5	43.3	23.2	5.8	8.2	9.1	9.8
$s^2 + s^3$	0.1	24.1	9.8	38.1	7.3	11.5	9.1
$\sin(\kappa s)$	0.1	37.7	0.6	35.4	2.8	19.8	3.6
$\tanh(\kappa s)$	0.2	37.9	1.3	30.9	5.4	17.3	7.0

Table 7: Significance tests for full-resolution CAVE completion at rank (33, 33, 4). Metrics are computed on missing entries.

ρ	RMSE				SAM			
	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

polynomial

$$h_\gamma(x) = \sum_{p=0}^4 \gamma_p x^p$$

by ridge regression on pairs $(\hat{Y}_{T,i}, Y_i)$. In full reconstruction, all entries are used for this fit. In completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted map is applied entrywise to $\hat{\mathcal{Y}}_T$; the Tucker core and factors are never updated.

MLPCal uses the same fixed-input protocol, but replaces the polynomial with a one-dimensional MLP. The input and target scalars are standardized, then a single-hidden-layer tanh network with a scalar skip connection is trained with Adam and ℓ_2 weight decay on the observed calibration pairs. In the CAVE runs we use 16 hidden units, learning rate 10^{-3} , weight decay 10^{-5} , batch size 8192, at most 200,000 calibration samples, and 1500 optimization steps. Thus both PolyCal and MLPCal test post-hoc scalar remapping of an already learned Tucker tensor, while NTD-PL jointly updates the latent Tucker geometry under the nonlinear link.

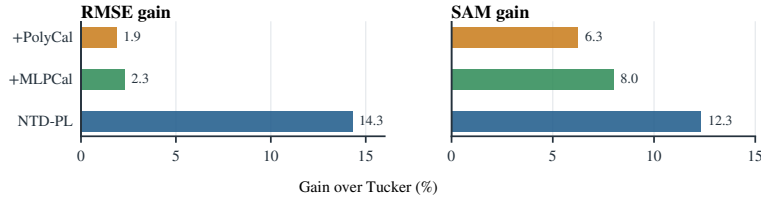


Figure 6: Post-hoc scalar calibration versus joint linked fitting on CAVE reconstruction at rank (33, 33, 4). Gains are relative to Tucker.

B.4 CAVE Beads Pseudo-RGB Visualization

Figure 7 visualizes the CAVE beads scene under the reconstruction and random-completion protocols. Each 31-band tensor is rendered as pseudo-RGB by selecting three representative spectral bands near 75%, 50%, and 20% of the wavelength index range and applying a shared linear intensity scaling within each task. For completion, unobserved entries in the observation panel are shown in gray.

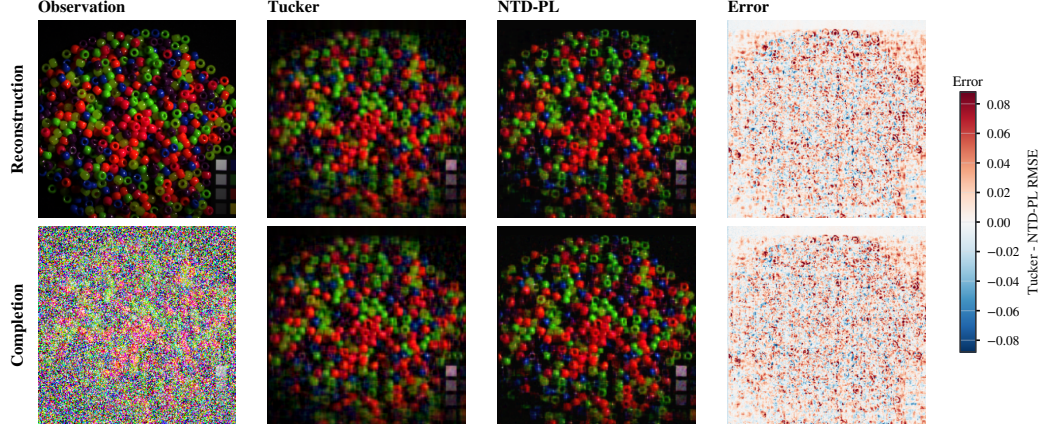


Figure 7: CAVE beads visual comparison. Rows correspond to reconstruction and random completion at missing rate 0.5. Error maps show Tucker minus NTD-PL per-pixel RMSE, so positive values indicate lower NTD-PL error.

B.5 Cross-Domain Real Tensor Checks

Table 8 reports matched-rank reconstruction on tensors outside the main hyperspectral setting: additional hyperspectral scenes, natural-image collections, and object-view tensors. We use NMSE(dB) as the main cross-dataset metric because it normalizes by the energy of each tensor and is therefore less sensitive to scale differences across domains. The results are consistent with the scoped claim in Section 6.4: gains are clearest on hyperspectral and object-view data, while natural-image collections provide more modest improvements.

Table 8: Additional matched-rank real tensor reconstruction. NMSE columns are in dB; lower is better. Δ NMSE is the Tucker-minus-NTD-PL dB gap, and NMSE gain is the relative reduction in linear NMSE; positive values favor NTD-PL.

Domain	Dataset	Shape	Rank	NMSE(dB)↓			NMSE gain
				Tucker	NTD-PL	Δ	
Hyperspectral	Jasper Ridge	$100 \times 100 \times 198$	(11,11,43)	-15.96	-16.58	0.62	13.4%
Hyperspectral	Samson	$95 \times 95 \times 156$	(10,10,35)	-19.37	-20.09	0.72	15.3%
Hyperspectral	Urban	$307 \times 307 \times 162$	(43,43,47)	-13.21	-13.53	0.32	7.1%
Natural images	CBSD68	$68 \times 96 \times 96 \times 3$	(20,32,32,3)	-11.27	-11.40	0.13	3.0%
Natural images	CIFAR-10	$1000 \times 32 \times 32 \times 3$	(96,20,20,3)	-17.04	-17.37	0.34	7.5%
Object views	COIL-100	$8 \times 36 \times 48 \times 48 \times 3$	(4,8,12,12,2)	-10.17	-10.71	0.54	11.7%
Object views	smallNORB	$25 \times 18 \times 64 \times 64 \times 2$	(18,12,24,24,2)	-25.97	-26.45	0.48	10.5%

NeurIPS Paper Checklist

This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed with the final 2026 submission template.

- Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment captions: NTD-PL is a parameter-efficient nonlinear observation model for Tucker tensors; its gains are not explained by modest rank increases or post-hoc scalar calibration.
- Limitations.** Section 7 discusses near-linear boundary cases, whole-axis missingness, shared-link assumptions, and the current domain concentration in hyperspectral data.
- Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial or analytic scalar links. The separation results use generic full-rank feature constructions. The masked generalization bound assumes bounded parameters, bounded observations, and random observed entries.

449 4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates,
450 and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured
451 missingness, and the cross-domain real-tensor reconstruction benchmark; the final version should
452 add hardware details and exact command manifests.